

Files

A web-based personal file manager — live at files.oppshnan.com.

by Warren Nocos for ITMD 504 — Programming and Application Foundations at Illinois Institute of Technology.

Users sign in with Google, upload, organize, download, rename, delete, and inspect files and folders through a browser-based interface. File content is encrypted at rest using AES/CTR with per-file IVs and a PBKDF2-derived key.

This is a one-page summary. For the full documentation, open `README.pdf` or read the pageless version of the [README on GitHub](#).

Quick links

Resource	URL
Live application	https://files.oppshnan.com
Source code	https://github.com/warrennocos/oppshnan-files
Full documentation	https://github.com/warrennocos/oppshnan-files/blob/main/README.md
CI/CD	https://github.com/warrennocos/oppshnan-files/actions
Figma prototype	https://figma.com/make/Wkr8DV1ZpKmnbnxNSMVgMs/Qppshnan-Files?p=f&fullscreen=1
Project board	https://github.com/users/warrennocos/projects/1

Tech stack

Layer	Stack
Backend	Java 25 · Virtual threads · Quarkus 3.34.3 · Hibernate ORM · Jakarta Data · Quarkus OIDC (Google)
Frontend	Angular 21 · TypeScript · SCSS · signals-first · RxJS · two-way data binding · custom event bus (CQRS)
Database	PostgreSQL 18 · Flyway migrations · recursive-CTE named native queries
Hosting	AWS EC2 t4g.small (Graviton 2 ARM) · Amazon Linux 2023 · Caddy TLS · Let's Encrypt · Route 53 DNS
CI/CD	Maven · GitHub Actions OIDC federation · AWS S3 and SSM Run Command (no SSH, no long-lived AWS keys)
Quality	JaCoCo coverage · Qodana static analysis · Quarkus Dev Services with real PostgreSQL via Testcontainers
Production binary	GraalVM 25 native image · ARM64

Project management

- 7 sprints on a GitHub Projects board
- 28 user stories with acceptance criteria, tracked as GitHub issues end-to-end
- One feature branch per epic, named `<issue#>-epic-XX-<slug>` (large epics ship in numbered parts; EPIC-07 spanned five)
- 24 wireframe mockups, one per user-facing screen state
- All commits reference issues via `refs #N <description>`

In this submission

- `README.pdf` — full 41-page documentation: architecture, data model, API reference, security, UX wireframes, CI/CD pipeline, development setup
- `README-summary.pdf` — this page